

GAYATRI KHARCHE

San Diego, CA | +1 858-518-9520 | gkharche@ucsd.edu | github.com/gayatrikharche | [linkedin.com/in/gayatri-kharche](https://www.linkedin.com/in/gayatri-kharche) | github.com/gayatrikharche

EDUCATION

UC SAN DIEGO

Master of Science in Machine Learning and Data Science

San Diego, CA

Sept 2023 - March 2025

Relevant Coursework: Statistical Learning, Data Analysis, Probability, Linear Algebra, Machine Learning, Deep Learning, Visual Learning, Big Network Data, Random Processes, Recommender System, Data Mining, Statistical NLP

SAVITRIBAI PHULE PUNE UNIVERSITY

Bachelor of Engineering in Electronics and Telecommunication

Pune, India

Aug 2019 - May 2023

GPA: 4.00/4.00

SKILLS

Technical Skills: Python, SQL, MATLAB, C++, PyTorch, TensorFlow, Keras, Pandas, NumPy, Seaborn, Matplotlib, Linux

Technologies: AI, Data Science, Data Analysis, Machine Learning, Deep Learning, Cloud Computing, Data Visualisation, GPU Programming, Tableau, AWS, Keras, LLM, Git, API, HTCondor, NLP, Docker, Kubernetes, SSH, RAG, CUDA

WORK EXPERIENCE

SAN DIEGO SUPERCOMPUTER CENTER

San Diego, CA

Machine Learning Student Researcher

Sept 2024 - Present

- Developing a RAG based LLM prototype for the SDSC Director's Office to streamline access to system usage data.
- Benchmarking ML models on OSPool, integrating PelicanFS for file transfer and comparing with previous methods.
- Conducting model ensembling to compare performance exploring methods to enhance AI and ML task efficiency.
- Deployed over 3 Docker containers to test Facebook's neural architecture search framework, gaining hands-on experience with containerized environments for scalable ML development.

SAN DIEGO SUPERCOMPUTER CENTER

San Diego, CA

Machine Learning Intern

July 2024 – Sept 2024

- Conducted performance benchmarking of across 10+ machine learning models with data ranging from 150 MB to 150 GB using PyTorch on OSPool and utilized HTCondor to efficiently submit and manage computational jobs.
- Tested and benchmarked GPU performance on low-dimensional time series modeling tasks using CopulaCPTS.
- Identified optimal data access methods on OSPool to enhance computational efficiency by 20%.

HEWLETT PACKARD ENTERPRISE

Pune, India

Software Intern

Feb 2023 – Aug 2023

- Trained in HPE GreenLake, hardware, cloud, networking, operating systems, VMware, and Linux fundamentals.
- Contributed to a VMware project by establishing domains, configuring virtual machines, and implementing DNS, NTP, and vCenter and utilized templates, content libraries, and migration strategies to optimize virtualized environment management.
- Improved virtual environment performance, resulting in a 10% increase in operational efficiency.
- Engineered scalable virtualized environments and gained hands-on experience in Windows Server Administration.

ORBITAL HOUSE PVT. LTD.

Pune, India

Project Intern

Dec 2021 – Jan 2022

- Played a pivotal role in a Home Automation project, utilizing ESP8266, cloud technologies, and various platforms.
- Integrated data analytics features to optimize energy usage, enhance user experience, and provide insights.

UNIVERSITY PROJECTS

HALLUCINATION DETECTION IN LLAMA 2 (7B)

Oct 2024 - Dec 2024

- Developed and evaluated hallucination detection models using transformer architectures (DistilBERT, ALBERT, BERT, and RoBERTa) on the Fever dataset, achieving around 70% accuracy on test set..
- Implemented an ensemble learning approach to improve model generalization and accuracy.

FIT PREDICTION USING RENT THE RUNWAY DATASET

Nov 2024 - Dec 2024

- Developed a recommendation system predicting clothing fit with 82.32% accuracy using a Text Mining-CNN model.
- Evaluated models like Logistic Regression, k-NN, and XGBoost and improved F2-scores by up to 20%.

IMAGE INPAINTING WITH CONTEXT ENCODERS

Jan 2024 - Mar 2024

- Implemented an auto-encoder for pixel context prediction, achieving a binary cross-entropy loss of ~ 0.54 .
- Trained Context Encoders for image inpainting, enhancing sharpness and demonstrating effective pre-training on CIFAR-10 with strong results.